

Four Bug Reports

Four support tickets, four hypotheses, and what happens when you look at the traces

Duration: 45 to 60 minutes

Group size: 3

Output: One Ticket Board per group plus a 75 second report out

Module context: Fine-Tuning and Retrieval-Augmented Generation. This discussion sits after the division of labor between retrieval and fine-tuning, and before the evaluation segment. You are being asked to diagnose without the metrics, on purpose.

READ THIS FIRST You have already seen these four symptoms once today, with a stage attached to each. Do not treat those as answers. Treat them as hypotheses that a one-line complaint from a frustrated user cannot possibly support. Your job is to work out what evidence would confirm or overturn each one, and then to look at that evidence.

1. How this session runs

You will work in a group of three. Three is deliberate. Nobody gets to be a passenger, every position must be held by someone, and a group of three cannot hide a disagreement behind a majority. In your roles, you will walk through and respond to each round (as described below).

2. Your role

Each person takes one role. Roles are not decoration. Each one owns a different definition of the word bad, and the friction between those definitions is where the useful part of this discussion happens. Argue your role even when you privately agree with someone else in the group.

Role	You own	The question you keep asking
Support operations lead	The ticket queue, the agents who file them, and the ones who stopped bothering	How many more of these are out there that nobody filed?
RAG engineer, on call	The chain, the index, the adapter, and whatever is deployed	Which stage, and what single piece of evidence would settle it?
Risk and resourcing partner	What ends up in a claim file, and what two engineers and a half-time SRE can finish before the freeze	Which of these do we have to answer for, and what are we giving up to fix it?

3. The system: Cordwell Assist

Cordwell Home and Hardware runs a support assistant called Cordwell Assist. It is the system you have been building today: a retriever over the support corpus, a small instruct model with the Cordwell LoRA adapter on top, and prompt assembly in between.

It reaches users two ways. The agent console is a widget inside the support tool your team owns. The Teams integration is a bot the store operations team built on top of your service, and they own that surface.

It has been in front of real users for five weeks. This week four tickets came in.

Stage	What it does today
Retrieve	Vector search over the Module 02 corpus. Top k of 4. Chunk size 1024 tokens, no overlap.
Assemble	System prompt, then the four retrieved chunks numbered and labeled with their source, then the user's question verbatim.
Generate	Small instruct model with the Cordwell LoRA adapter, which enforces a JSON schema with an answer field, a citations list, and a confidence field.

4. Environment facts

Established fact for the duration of this discussion. Do not invent facts that contradict it. Do notice what is absent.

Item	Value
Base model	Held in BASE_MODEL. A small instruct model, roughly 360M parameters.
Adapter	cordwell, a LoRA adapter. Trained on 1,200 examples, of which 340 (28 percent) are abstention examples where the correct output is a refusal.
Retriever	Top k of 4 against the Module 02 index.
Surfaces	Agent console service, owned by your team. Teams integration, owned by store operations.
Volume	About 2,400 assisted sessions per week. Roughly 70 percent agent console, 30 percent Teams.
Ticket queue	4 tickets this week. 7 last week. Nobody has read last week's.
Deployments in the last seven days	Monday: adapter update. Wednesday: prompt template edit. Thursday: index refresh. All three shipped by different people.
Monitoring in place	Latency percentiles per request, and HTTP error rate.
Quality monitoring	None.
Team	Two engineers and a half-time SRE.
Calendar	The retail change freeze starts in six weeks.

Cordwell Home and Hardware is fictional. These figures are synthetic and internally consistent. Treat them as real for the next hour.

5. What gets logged, and what does not

This is the most important table in the packet. Read it twice.

Recorded for every request	Not recorded at all
Timestamp	Which chunks were retrieved
Surface (agent console or Teams)	The retrieval scores
Query text as typed	Whether the adapter was enabled
Answer text as returned	Temperature and sampling settings
Latency	Which version of the system prompt was used
HTTP status	Which base model version was loaded

6. The four tickets

These are the tickets as filed, plus what the ticket does and does not contain. The quoted lines are what the reporter wrote. Everything under "also in the ticket" is what the engineer who triaged it added.

Ticket 1 | CA-2291

Filed by Dana R., support agent. Agent console. Tuesday 09:41.

"It told me the thermostat needs a 40-amp circuit. It needs 15. The manual says 15."

Also in the ticket

- Dana attached a photo of page 4 of the printed TH-40 installation manual. It says 15 amps.
- Dana did not record the exact question she typed.
- The customer at the other end of that call was an electrician preparing to pull a permit.

Ticket 2 | CA-2294

Filed by store operations on behalf of an unnamed associate. Reported by phone. Tuesday 14:02.

"It said it couldn't find anything about wifi pairing. There's a whole troubleshooting doc about it."

Also in the ticket

- The troubleshooting document does exist in the CMS.
- The ticket does not record which surface the associate was using.
- The ticket does not record the exact question, only that it was about wifi pairing.

Ticket 3 | CA-2301

Filed by Marcus T., support agent. Agent console. Wednesday 11:15.

"The answer was right, but it cited a document about garage doors."

Also in the ticket

- Marcus checked the answer against the manual himself. It was correct.
- Marcus did not check whether the cited document was among the retrieved sources, because the agent console does not display retrieved sources.
- Marcus filed this as low priority, on the grounds that the answer was right.

Ticket 4 | CA-2288, CA-2296, CA-2299, CA-2305

Filed by four different store associates. All four via the Teams integration. Across four days.

"Every answer starts with 'Certainly! I'd be happy to help.' We asked it to stop four times."

Also in the ticket

- The phrase "we asked it to stop four times" refers to four messages typed into the Teams thread telling the bot to stop.
- No change was made to any prompt, any configuration, or any model between the four reports.
- None of the four associates reported a problem with the content of the answers.

7. Round 0: your first instinct

Silent. Individual. Do not discuss.

Write one line per ticket: the stage you think broke. Then write, next to it, how confident you are on a scale of one to five.

That confidence number is the point of this round. You will look at it again after the traces land, and the gap between your confidence and your accuracy is the single most useful thing this session can show you.

Ticket	The stage I think broke	Confidence 1 to 5	Was I right
1			
2			
3			
4			

8. Round 1: what would you need to see

Group. You are not diagnosing yet.

This round has one rule and it is strict: no fixes and no conclusions. You are building an evidence request.

- 1.1** For each ticket, name the single artifact you would need to see to confirm or overturn your Round 0 answer. An artifact is a specific thing, not a category. "The logs" does not count. "The retrieved chunk IDs and their scores for that exact query" does.

- 1.2 For each ticket, separate what the reporter observed from what the reporter inferred. At least two of these tickets contain an inference wearing the clothes of a fact. Find them and say what the inference is.
- 1.3 One ticket's stated evidence does not support its stated conclusion at all, and you can prove that from the ticket text alone, before you see a single trace. Which one, and what is the gap?
- 1.4 Which of these four is dangerous rather than merely wrong? Name what makes it dangerous and say whether that changes what you do first or only what you say to whom.

9. Round 2: re-diagnose against the traces

Group. Read all four traces in the "Handout B" section before anyone speaks.

Fill the Ticket Board in Section 10 as you go. A diagnosis without a line of evidence attached to it is still a hypothesis.

- 2.1 For each ticket, state three things: the stage, the defect stated precisely, and the one line in the trace that proves it. "Generation failure" is not a defect stated precisely. "The model refused to answer despite the answer being in the retrieved set at rank one" is.
- 2.2 Compare against your Round 0 answers. Two of the four changed stage outright. One turned out to be two failures at once, where the more visible one concealed the other. One is not a model problem at all. Sort them into those buckets, then answer the harder question: what does that tell you about how much diagnostic weight a one-line symptom can carry?
- 2.4 The fix for Ticket 2 and the fix for Ticket 3 pull in opposite directions. Say exactly how. Then name the pair of numbers you would have to watch to know you had not traded one failure for the other. One number is not enough and saying why is most of the answer.
- 2.5 Ticket 4's trace shows that the adapter your team trained is not loaded in the service the users hit. Before anyone writes any code: what do you check, and what does that fact do to your confidence in the other three diagnoses?
- 2.6 One of these four cannot be diagnosed to a root cause from its trace, only to a stage. Which one, what is missing, and what do you do with a ticket you cannot finish diagnosing? "Close it" is an answer you may give if you can defend it.
- 2.7 You get one deployment this week. What is in it, and what single measurement do you run before and after to know it worked? Name the number and the direction you expect it to move. If you cannot name a number, you cannot know.

10. Your deliverable: the Ticket Board

One per group. Hand it to your facilitator at the end.

Part 1. Diagnosis

Ticket	Stage	The defect, precisely	The line that proves it
1			

Ticket	Stage	The defect, precisely	The line that proves it
2			
3			
4			

Part 2. Action

Ticket	The fix	Cost: free, hours, or days	Order	The number you watch, and which way it should move
1				
2				
3				
4				

11. Round 3: report out

#	State this	What makes it count
1	Which ticket you fix first, and the one line of the trace that decided it	Quote the line. Not the stage, not the reasoning, the line.
2	The one thing you refuse to do, and the single fact that would change your mind	Name the fact and name who has it. "More information" is not a fact.
3	Where your group genuinely disagreed, and whether you resolved it	Not optional. In a group of three there is no majority, so a clean consensus means somebody gave in.
4	The one automated check you would add so this class of ticket never reaches a human again	Name it precisely enough that someone could build it Monday. "Better monitoring" is not a check.

12. Vocabulary for this discussion

Term	Working definition in this context
Stage attribution	Deciding whether a failure happened in retrieval or in generation. Everything downstream of that decision depends on getting it right.
Grounded	Every claim in the answer is supported by something in the retrieved context. An answer can be correct and ungrounded at the same time, and that is worse than it sounds.
Faithfulness	The degree to which an answer is supported by its retrieved context. Not the same as whether the answer is true.
Abstention	The model declining to answer. It has two error directions: refusing when it could have answered and answering when it should have refused. Fixing one can worsen the other.
Citation verification	Checking that every source the model cited was in the retrieved set. Cheap, and it catches a real failure mode.
Reconstruction	A trace produced by re-running the query later. It shows what the system does now, not what it did then.
Configuration drift	The deployed configuration differing from the one you tested against. It produces failures that look like model problems and are not.
One variable at a time	If three things shipped this week and quality moved, you cannot attribute the movement. This is why deploy discipline is a diagnostic tool, not a bureaucratic one.

The Trace Pack

Cover note from the engineer who pulled these

- Traces for Tickets 1, 2, and 3 were produced by re-running the reported query today. They are reconstructions, not recordings. Section 5 explains why no recording exists.
- The trace for Ticket 4 is a recording, taken from a debug session that happened to be running on the Teams service on Thursday.
- For Tickets 1 and 3 the surface is known, because the ticket names the agent console. For Ticket 2 it is not recorded.
- Scores are cosine similarity. Top k is 4. Only the chunks relevant to the complaint are shown in full.

Trace: Ticket 1 (CA-2291)

```

SURFACE: agent console      ADAPTER: enabled      TEMPERATURE: 0.1

QUERY (reconstructed from Dana's description):
  "breaker size for TH-40"

RETRIEVED (top 4):
[1] install_manual_th40.md#12    score 0.79
   "...the panel requires a dedicated 40 amp circuit. Terminate the
   feed at the load center before mounting. Torque lugs to 35 in-lb..."
[2] install_manual_th40.md#11    score 0.74
   "...sold separately and required for multi-zone installations.
   7. Zoning Panel Accessory (ZP-4)"
[3] faq_thermostats.md#2        score 0.68
[4] install_manual_th40.md#3     score 0.61

ANSWER:
{"answer": "The TH-40 requires a dedicated 40 amp circuit.",
 "citations": ["install_manual_th40.md#12"], "confidence": "high"}

ENGINEER'S NOTE:
Chunk #12 is the body of Section 7. The heading
"7. Zoning Panel Accessory (ZP-4)" is the LAST line of chunk #11.
Chunk size 1024, no overlap. The 40 amp figure is real and correct
for the ZP-4 panel. The TH-40 thermostat itself is low voltage.
```

Trace: Ticket 2 (CA-2294)

```

SURFACE: not recorded      ADAPTER: not captured  TEMPERATURE: not captured

QUERY (reconstructed):
  "how do I pair the wifi on a TH-40"

RETRIEVED (top 4):
[1] troubleshoot_wifi.md#7       score 0.81
   "...to pair, hold the MODE button for five seconds until the LED
   flashes amber, then select CORDWELL-TH40 in the Cordwell app..."
[2] troubleshoot_wifi.md#8       score 0.77
[3] faq_connectivity.md#1        score 0.71
```



```
[4] install_manual_th40.md#3      score 0.64
```

ANSWER:

```
{"answer": "I could not find information about wifi pairing in the  
provided documentation.", "citations": [], "confidence": "low"}
```

ENGINEER'S NOTE:

Re-running this query today reproduces the refusal about half the time. The other half it answers correctly from chunk #7.

Trace: Ticket 3 (CA-2301)

SURFACE: agent console ADAPTER: enabled TEMPERATURE: 0.1

QUERY (reconstructed):

"how high should the TH-40 be mounted"

RETRIEVED (top 4):

```
[1] install_manual_th40.md#2      score 0.66  
    "...remove the existing wall plate and label each conductor  
    before disconnecting..."  
[2] faq_thermostats.md#5         score 0.63  
[3] install_manual_th40.md#9      score 0.61  
[4] warranty_terms.md#1          score 0.58
```

No retrieved chunk contains a mounting height.

ANSWER:

```
{"answer": "Mount the TH-40 approximately 52 to 60 inches above the  
finished floor, away from direct sunlight and supply registers.",  
"citations": ["garage_opener_gd200.md#4"], "confidence": "high"}
```

ENGINEER'S NOTE:

garage_opener_gd200.md#4 was NOT in the retrieved set.
The answer is correct. The corpus does contain the mounting height,
at install_manual_th40.md#5, which was not retrieved for this query.

Trace: Ticket 4 (CA-2288 and the three that followed)

SURFACE: Teams integration (this one is a RECORDING)

QUERY:

"what's the warranty on the TH-40"

ANSWER (verbatim, not JSON):

"Certainly! I'd be happy to help. Based on the documentation, the TH-40 carries a two-year limited warranty on parts..."

SERVICE CONFIG AT TIME OF REQUEST (Teams integration):

```
BASE_MODEL      = <the small instruct model>  
ADAPTER_PATH    = (unset)  
temperature     = 0.9  
system prompt   = "You are a helpful assistant."
```

NOTEBOOK CONFIG (where your team tests):

```
ADAPTER_PATH    = ./adapters/cordwell  
temperature     = 0.1  
system prompt   = the Cordwell support system prompt
```

ENGINEER'S NOTE:

All four reports in this ticket group came through Teams.
The agent console service is a separate deployment.
Nobody has checked which configuration the agent console is running.

BEFORE YOU DIAGNOSE Read all four traces first. At least one of them is easier to misread on its own than it is next to the other three.